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Kelas: TI-20

$$1) \int_0^1 \frac{dx}{(2x-1)^3} \quad x_0 = 0 \quad h = 0.2$$

$$x_n = 1$$

Integral :  $-\frac{1}{4(2x-1)^2}$

Eksak :  $-\frac{1}{4(2 \cdot 1 - 1)^2} - \left(-\frac{1}{4(2 \cdot 0 - 1)^2}\right)$

$$= 0 //$$

|    |    |          |      |     |         |   |
|----|----|----------|------|-----|---------|---|
| x  | 0  | 0.2      | 0.4  | 0.6 | 0.8     | 1 |
| fx | -1 | -4.62963 | -125 | 125 | 4.62963 | 1 |

⇒ Riemann

$$h \sum_{i=0}^n f(x_i)$$

$$= 0.2 (-1 + (-4.62963) + (-125) + 125 + (4.62963) + 1)$$

$$= -8.0646$$

$$\text{error} = 0 - (-8.0646) = 8.0646 //$$

⇒ Trapezoid

$$\frac{h}{2} (f_0 + 2 \sum_{i=1}^{n-1} f_i + f_n)$$

$$= \frac{0.2}{2} (-1 + (-8.0646) + 1)$$

$$= -8.0646$$

$$\text{error} = 0 - (-8.0646) = 8.0646 //$$

$\Rightarrow$  Simson  $\frac{1}{3}$

$$\frac{h}{3} \left( f_0 + \underset{1 \text{ ganjil}}{4 \sum f_i} + \underset{1 \text{ genap}}{2 \sum f_i} + f_n \right)$$

$$= 16.04938$$

$$\text{error} = 0 - 16.04938 = -16.04938$$

Simson  $\frac{3}{8}$

$$\frac{3h}{8} (f_0 + 3f_1 + 3f_2 + 2f_3 + f_n)$$

$$= -9.375$$

$$\text{error} = 0 - (-9.375) = 9.375$$

3)  $5x^3 - 6x + 12x^{-2}$   $x_0 = 1$   $h = 0.2$   
 $x_n = 3$

$$\text{Integral} = \frac{5}{3} x^3 - 3x^2 - \frac{12}{x}$$

$$\text{Exact} = \left( \frac{5 \cdot 3^3}{3} - 3 \cdot 3^2 - \frac{12}{3} \right) - \left( \frac{5 \cdot 1^3}{3} - 3 \cdot 1^2 - \frac{12}{1} \right) = 27.333$$

$\Rightarrow$  Reiman

|      |    |       |        |        |        |   |        |        |       |        |
|------|----|-------|--------|--------|--------|---|--------|--------|-------|--------|
| x    | 1  | 1.2   | 1.4    | 1.6    | 1.8    | 2 | 2.2    | 2.4    | 2.6   | 2.8    |
| f(x) | 11 | 8.333 | 7.5224 | 7.0023 | 6.1032 | 4 | 13.929 | 16.403 | 19.92 | 22.923 |

$$L = h \sum_{i=0}^{n-1} f(x_i)$$

$$= 31.40925$$

$$\text{error} = 27.333 - 31.40925$$

$$= -4.076417$$

⇒ Trapezoid

$$\frac{h}{2} (f_0 + 2 \cdot \sum_{i=1}^{n-1} f_i + f_n)$$

$$= 27.42642$$

$$\text{error: } 27.333 - 27.426421 = 0.143089$$

⇒ Simpson  $\frac{1}{3}$

$$\frac{h}{3} (f_0 + 4 \sum_{i=1}^{n-1} f_i + 2 \sum_{i=2}^{n-2} f_i + f_n)$$

$$= 27.3355$$

$$\text{error: } 27.3333 - 27.3355 = 0.002269$$

⇒ Simpson  $\frac{3}{8}$

$$\frac{3h}{8} (f_0 + 3f_1 + 3f_2 + 2f_3 + \dots + f_n)$$

$$= 26.03866$$

$$\text{error: } 27.3333 - 26.03866 = 1.294623$$

1).  $(2x^2 - 3)^2$   $x_0 = -1$   $h = 0.3$   
 $x_n = 2$

$$\text{Integral: } \frac{4x^5}{5} - 4x^3 + 9x$$

$$\text{F.S. at: } \left( \frac{4 \cdot 2^5}{5} - 4 \cdot 2^3 + 9 \cdot 2 \right) - \left( \frac{4 \cdot (-1)^5}{5} - 4 \cdot (-1)^3 + 9 \cdot (-1) \right)$$

$$= 17.9$$

|      |    |        |        |        |        |      |        |        |      |      |    |
|------|----|--------|--------|--------|--------|------|--------|--------|------|------|----|
| x    | -1 | -0.7   | -0.4   | -0.1   | 0.2    | 0.5  | 0.8    | 1.1    | 1.4  | 1.7  | 2  |
| f(x) | 1  | 9.0804 | 2.1024 | 8.8804 | 8.5204 | 6.25 | 2.9504 | 0.3364 | 0.84 | 2.28 | 25 |

⇒ Riemann

$$h \sum_{i=0}^n f(x_i)$$

$$= 21,83626$$

$$\text{error} = 17,4 - 21,83626 = 4,43626 //$$

⇒ trapezoid

$$\frac{h}{2} (f_0 + \sum_{i=1}^{n-1} f_i + f_n)$$

$$= 17,93676$$

$$\text{error} = 17,4 - 17,936261 = 0,53626 //$$

⇒ Simpson  $\frac{1}{3}$

$$\frac{h}{3} (f_0 + \underset{\text{Ganjil}}{4 \sum f_i} + \underset{\text{Genap}}{2 \sum f_i} + f_n)$$

$$= 17,41296$$

$$\text{error} = 17,4 - 17,41296 = 0,01296 //$$

Simpson  $\frac{3}{8}$

$$\frac{3h}{8} (f_0 + 3f_1 + 3f_2 + 2f_3 \dots f_n)$$

$$= 16,51505$$

$$\text{error} = 17,4 - 16,51505 = 0,88495 //$$